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Step 12 - CNTVOFF_EL2

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture documentation.

Previous chapters introduced:

- Physical Counter (`CNTPCT_EL0`)
- Virtual Counter (`CNTVCT_EL0`)
- Virtual Timer

This chapter explains the register that makes timer virtualization possible:

CNTVOFF_EL2

Without `CNTVOFF_EL2`, the Virtual Counter would simply mirror the Physical Counter.

Learning Objectives

After completing this chapter, you should understand:

- Purpose of `CNTVOFF_EL2`
- How virtual time is created
- Relationship between Physical and Virtual counters
- Why offset-based virtualization is efficient
- Guest startup scenarios
- VM migration concepts
- Practical offset calculations

1. What is CNTVOFF_EL2?

`CNTVOFF_EL2` is the:

Virtual Counter Offset Register

Owned and managed by:

EL2 (Hypervisor)

Its purpose is to create:

Virtual Time

from:

Physical Time

2. Why is CNTVOFF_EL2 Needed?

Without an offset:

Register	Value
CNTPCT	1,000,000
CNTVCT	1,000,000

Result:

Physical Time = Virtual Time

No virtualization exists.

Desired Behavior

Suppose a Guest VM starts now.

Physical counter:

1,000,000

Guest should see:

0

This requires a transformation.

ARM performs:

Virtual Time
=
Physical Time - Offset

3. Core Formula

The most important formula in ARM timer virtualization:

CNTVCT
=
CNTPCT - CNTVOFF_EL2

Where:

Symbol	Meaning
CNTPCT	Physical Counter
CNTVOFF_EL2	Virtual Offset
CNTVCT	Virtual Counter

4. Example - No Virtualization

Parameter	Value
CNTPCT	1,000,000
CNTVOFF_EL2	0

Calculation:

CNTVCT
=
1,000,000 − 0
=
1,000,000

Result:

Counter	Value
Physical Counter	1,000,000
Virtual Counter	1,000,000

The counters are identical.

5. Example - Guest Starts at Zero

Suppose:

Parameter	Value
CNTPCT	1,000,000
Desired Guest Time	0

Hypervisor programs:

CNTVOFF_EL2
=
1,000,000

Calculation:

CNTVCT
=
1,000,000 − 1,000,000
=
0

Guest observes:

Time = 0

even though hardware has already been running.

6. Example - Guest Starts Later

Suppose:

Parameter	Value
CNTPCT	1,000,000
Desired Guest Time	500,000

Calculate:

CNTVOFF_EL2
=
1,000,000 − 500,000
=
500,000

Result:

CNTVCT
=
1,000,000 − 500,000
=
500,000

7. Visualizing the Offset

View	Counter Value
Physical Counter	1,000,000
Offset	250,000
Virtual Counter	750,000

Conceptually:

Physical Timeline
---------->
Offset

↓

Virtual Timeline

---------->

The offset shifts the visible timeline.

8. Why ARM Uses Offsets

ARM could have created:

Option	Cost
One hardware counter per VM	High
One shared System Counter + Offset	Low

ARM chose:

One System Counter

+

Offset Register

Benefits:

Benefit	Description
Lower hardware cost	No duplicate counters
Scalability	Supports many guests
Simplicity	Reuses existing timer hardware
Performance	No software emulation required

9. VM Migration Example

Suppose a guest currently sees:

Guest Time = 10 seconds

Physical machine A:

Item	Value
Physical Counter	1,000,000,000
Guest Time	10 seconds

When moved to Machine B:

The hypervisor can program an appropriate:

CNTVOFF_EL2

so the guest continues to see:

Guest Time = 10 seconds

The guest's timeline remains continuous.

10. Relationship with Virtual Timer

Remember:

Component	Source
Physical Timer	CNTPCT
Virtual Timer	CNTVCT

Since the Virtual Timer uses:

CNTVCT

changing:

CNTVOFF_EL2

also changes:

Virtual Timer behavior

This is how ARM virtualizes timer events.

Architecture Summary

Component	Purpose
System Counter	Real hardware time
CNTPCT_EL0	Physical view of time
CNTVOFF_EL2	Virtual offset
CNTVCT_EL0	Virtual view of time
Virtual Timer	Uses virtual time

Common Misconceptions

Misconception 1

CNTVOFF_EL2 changes the System Counter

False.

The System Counter never changes.

Only the visible virtual view changes.

Misconception 2

Virtual Counter is another hardware counter

False.

It is derived from:

CNTPCT − CNTVOFF_EL2

Misconception 3

Offset affects Physical Time

False.

Only Virtual Time is affected.

Key Takeaways

Concept	Key Point
CNTVOFF_EL2	Virtual Counter Offset
Managed By	EL2
Purpose	Create virtual time
Formula	CNTVCT = CNTPCT − CNTVOFF_EL2
Physical Counter	Unchanged
Virtual Counter	Offset-adjusted
Hardware Cost	No extra counters required

Knowledge Check

Q1

What is the purpose of `CNTVOFF_EL2`?

Q2

What is the formula for calculating `CNTVCT`?

Q3

Does `CNTVOFF_EL2` modify the System Counter?

Q4

Who typically programs `CNTVOFF_EL2`?

Q5

Why does ARM use an offset instead of creating another hardware counter?

Answer Key

Question	Answer
Q1	Create virtualized time
Q2	CNTVCT = CNTPCT − CNTVOFF_EL2
Q3	No
Q4	Hypervisor (EL2)
Q5	Lower cost and simpler hardware